CURRICULUM VITAE

1. Personal information

Full name	: **NGUYEN THI KIM ANH**
Birth date	: May 16th, 1993
Marital status	: Single
Email	: ntkanh165@gmail.com
Nationality	: Vietnam

2. Academic backgrounds

Ph. D. in Biomedical Engineering and Environmental Sciences

National Tsing-Hua University, Taiwan (2019 – 2023)

Thesis title: *Enhanced electrochemical performance of carbon-based nanocomposites for advancements in capacitive deionization applications. (Supervisor: Prof. Ruey-an Doong)*

MSc. in Environmental Engineering

National Central University, Taiwan (2017 – 2019)

Thesis title: *Electro-oxidation of acid orange 7 by Ti-PbO2 modified activated carbon electrode. (Supervisor: Prof. Ching-Ju Monica Chin)*

BSc. in Environment Science

Sai Gon University, Vietnam (2011 – 2015)

Thesis title: *Removal organic pollutants from hospital wastewater using gamma ray irradiation method. (Supervisor: Prof. Bui Manh Ha)*

3. Employment experiences

Postdoctoral fellowship July 2023 – now

Division of Hematology-Oncology, Department of Internal Medicine, Chang Gung Memorial Hospital (Taiwan).

Student researcher August 2015 – December 2016

Department of Environmental Sciences, Sai Gon University.

Industrial engineering at Nobland Vietnam Co., Ltd (Vietnam).

Internship June 2014 – October 2014

Sub-institute of hydrometeorology and climate change (Vietnam)..

4. Achievements

- International scholarship for Ph.D. program (2019-2023) in Department of Biomedical Engineering and Environmental Sciences (BMES), National Tsing-Hua University (Taiwan).
- International scholarship for Master program (2017-2019) in Graduate Institute of Environmental Engineering, National Central University (Taiwan).
- Excellent Student of the year (2012) from Department of Environmental Sciences, SaiGon University (Vietnam).

5. Research interest

- Development of nanomaterials for energy, environmental, and biomedical applications: Metal-organic frameworks, MXene, carbon-based materials, and transition metal dichalcogenides.
- Water treatment and energy nexus applications.
- Electrochemical biosensing devices for cancer diagnosis.

6. Research project (2019-now)

- Visible-light responsive photoelectrochemical reaction of metal oxide/2-dimensional porous materials for pollutants degradation and energy recovery (**MOST107-2221-E007-113-MY3**)
- Taiwan-Malaysia clean water quality and sustainable energy overseas research (**MOST 108-2911-I-007-302**)
- Upconversion of graphene quantum dots/porous semiconductor nancomposites for nanobiosensing and energy conversion and storage (**MOST108-2113-M007-021-MY3**)
- Development of 2-Dimensional layered mxene-based photoeletrochemical materials for water-energy and net zero emission platform applications (**NSTC112-2223-E007-010**)
- Metal organic frameworks/covalent organic framework composites for nanozyme sensing and energy conversion and storage applications (**NSTC113-2113-M007-003**)
- Electrochemical biosensor based on lectin-functionalized nitrogen, sulfur-doped graphene quantum dot decorated gold nanoparticles for breast cancer diagnosis: from academic research to clinical translation (**CGMH-CORPG3P0391**)

7. Publications

[1] **T. K. A. Nguyen**, V. T. Nguyen, T.-V. Huynh, R.-A. Doong, and J. C.-H. Hsieh, Constructing an ultrasensitive electrochemical sensor based on N, S-codoped graphene quantum dots decorated Au-nanostar for MCF-7 breast cancer diagnostic, Chemical Engineering Journal (2025). (**IF: 13.2**)

[2] T.-V. Huynh, **T. K. A. Nguyen**, J. C.-H. Hsieh, and R.-A. Doong, Electrochemical sensor for efficient cancer biomarker CA 125 diagnostic, prognostic, and prediction of pancreatic ductal adenocarcinoma (PDACs) using magnetically porous carbon-integrated Er-GQDs, Microchemical Journal (2025). (**IF: 5.1**)

[3] I. H. Chiang , H.-C. Kuo, C.-T. Liao, Y.-C. Kuo, S.-M. Yu, H.-M. Wang, Y.-H. Huang, **T. K. A. Nguyen**, M.-H. Wu, and J. C.-H. Hsieh, A two-dimensional Ex-vivo cultivation model for circulating tumor cells: high success rate and correlations with cancer response to therapies, Biomedical Journal 48 (2025) 100891. (**IF: 4.4**)

[4] N. T. N. Anh, T.-V. Huynh, V. T. Nguyen, **T. K. A. Nguyen**, M.-H. Wu, and J. C.-H. Hsieh, MXene nanosheet-derived N, S-codoped graphene quantum dots for ultrasensitive and selective detection of 3-nitro-l-tyrosine in human serum, Analytica Chimica Acta 1292 (2024) 342237. (**IF: 6.0**)

[5] **T. K. A. Nguyen**, N. T. N. Anh, M. D. Nguyen, V. T. Nguyen, and R.-a. Doong, Boosting capacitive deionization of monovalent and hardness ions using $Ti_3C_2T_x$ MXene as an intercalation-type pseudocapacitive electrode, Separation and purification technology, 327 (2023) 124934. (**IF: 9.0**)

[6] **T. K. A. Nguyen**, T.-V. Nguyen, and R.-a. Doong, Enhanced capacitive deionization of Cr(VI) using functionalized metal carbide 2D framework and badam tree leaf-derived carbon as the asymmetric electrode materials, Chemical Engineering Journal, 475 (2023) 146439. (**IF: 13.2**)

[7] **T. K. A. Nguyen**, T.-H. Wang, and R.-a. Doong, Architectures of flower-like MoS2 nanosheet coated N-doped carbon sphere electrode materials for enhanced capacitive deionization, Desalination 540 (2022) 115979. (**IF: 9.8**)

[8] **T. K. A. Nguyen**, E. P. Kuncoro, and R.-A. Doong, Manganese ferrite decorated N-doped polyacrylonitrile-based carbon nanofiber for the enhanced capacitive deionization, Electrochimica Acta 401 (2022) 139488. (**IF: 5.6**)

[9] T. Ketwong, E. Rabang Halabaso, **T. K. A. Nguyen**, C. Areeprasert, and R.-A. Doong, Comparative study on pilot-scale production of CuO-loaded activated biochar and hydrochar from oil-palm empty fruit bunches for high-performance symmetric supercapacitor application, Journal of Electroanalytical Chemistry 905 (2022) 115970. **(IF: 4.6**)

[10] T.-H. Wang, **T. K. A. Nguyen**, and R.-a. Doong, Phosphorene nanosheet decorated graphitic carbon nitride nanofiber for photoelectrochemically enhanced hydrogen evolution from water splitting, Journal of the Taiwan Institute of Chemical Engineers 141 (2022) 104577. (**IF: 6.3**)

[11] **T. K. A. Nguyen**, Y.-H. Huang, J. C.-H. Hsieh, Comprehensive review of electrochemical biosensors for sensitive detection circulating tumor cells in blood: advances and challenges. (***to be sumitted***)

[12] **T. K. A. Nguyen**, V. T. Nguyen, M. D. Nguyen, and R.-A. Doong, Synthesis of 2D/2D $MoS_2/Ti_3C_2T_x$ MXene heterostructure for high performance supercapacitor. (***to be sumitted***)

[13] V. T. Nguyen, **T. K. A. Nguyen,** R. A. Doong, Unleashing the potential of capacitive deionization via surface engineering of Ti_3C_2 MXene nanosheet by Mo_{132} (Mo_{132}-MX) for ammonium capture. (***to be submitted***)

8. Conference talks

- November 26th – 28th, 2020: Oral presentation at **The 2020 International Conference on Green Electrochemical Technologies (2020 ICGET-Tw)** with **"**Hierarchical zeolitic imidazolate framework-derived manganese ferrite decorated activated carbon fiber for the improvement of capacitive deionization performance**".**
- November 5th, 2021: Award for the best oral presentation in the **Conference of "The 18th Cross-Strait Academic Conference on Carbon Materials and the 5th annual Carbon Society of Taiwan"** with "Enhanced capacitive deionization performance by using MoS_2 nanosheets coated N-doped-carbon-sphere". ***Best oral presentation award***

- November 12th, 2022: Oral presentation at **The 2022 International Conference on Green Electrochemical Technologies (2022 ICGET-Tw)** with **"**Effective removal of Cr(VI) using the badam tree leaf-derived carbon and functionalized metal carbide 2D framework for asymmetric CDI electrode".
- April 10th – 14th, 2023**:** Oral presentation at **Material Research Society Spring Meeting and Exhibit** with "Synthesis of 2D/2D $MoS_2/Ti_3C_2T_x$ MXene heterostructure for high performance supercapacitor".
- May 30th – June 3rd, 2023: Oral presentation at **The 18th IWA Leading Edge Conference on Water and Wastewater Technologies Achieving Sustainability Through Water Technology** with "Enhanced Asymmetric Capacitive Deionization Using Fe-MOF/rGO And GQDs@PANI/rGO Nanocomposites As Electrode Materials".
- July 2nd – 6th, 2023: Oral presentation at **The 6th International Conference on Battery Deionization & Electrochemical Separation** with "MXene $Ti_3C_2T_x$ as an intercalation-type pseudocapacitive electrode for boosting capacitive deionization performance".
- December 2nd – 8th, 2024: Poster presentation at **Material Research Society Fall Meeting and Exhibit with "**Electrochemical biosensor based on lectin-functionalized nitrogen, sulfur-doped graphene quantum dot decorated gold nanostars for Breast Cancer Diagnosis: From academic research to clinical translation**".**

9. Workshop and Training

- November 5th – 16th, 2019: Completion of the **2019 Southeast Asia International Joint Research and Training Program on Environmentally Green Technology and Low Carbon Energy.**
- February 4th – 15th, 2020: Completion of the **2020 Southeast Asia International Joint Research and Training Workshop on Environmentally Green Technology and Sustainable Energy.**
- November 29th – December 5th, 2020: Completion of the **Southeast Asia International Joint Research and Training Program on Sustainable Development.**

- August 2nd – 12th, 2021: Completion of the **Workshop on Circular Economy Based Environmentally Green Technology for Sustainable Development.**

10. Professional skills

- **Material characterization techniques:**

  Instrument licenses: XRD, BET, XAS, FE-SEM, FTIR, and Raman spectra.

- **Sensing and energy related applications:**

  Instrument licenses: Electrochemical instruments (Autolab, CHI, and Biologic), Ion chromatography (IC), HPLC, UV-visible.

- **Data analysis software**: Microsoft Office™ tools (Word, Excel, PowerPoint), SPSS, Originlab, XPSpeak..

Declaration

I declare that the information supplied by me on this form is complete, true and correct in every particular. I am aware that there are severe penalties for providing false or misleading information.